Padrões de Diversidade de Espécies

Tópicos em Zoologia 1

UnB - Depto. De Zoologia
Roberto B. Cavalcanti - 24 de março de 2026

Padrão Log-Normal

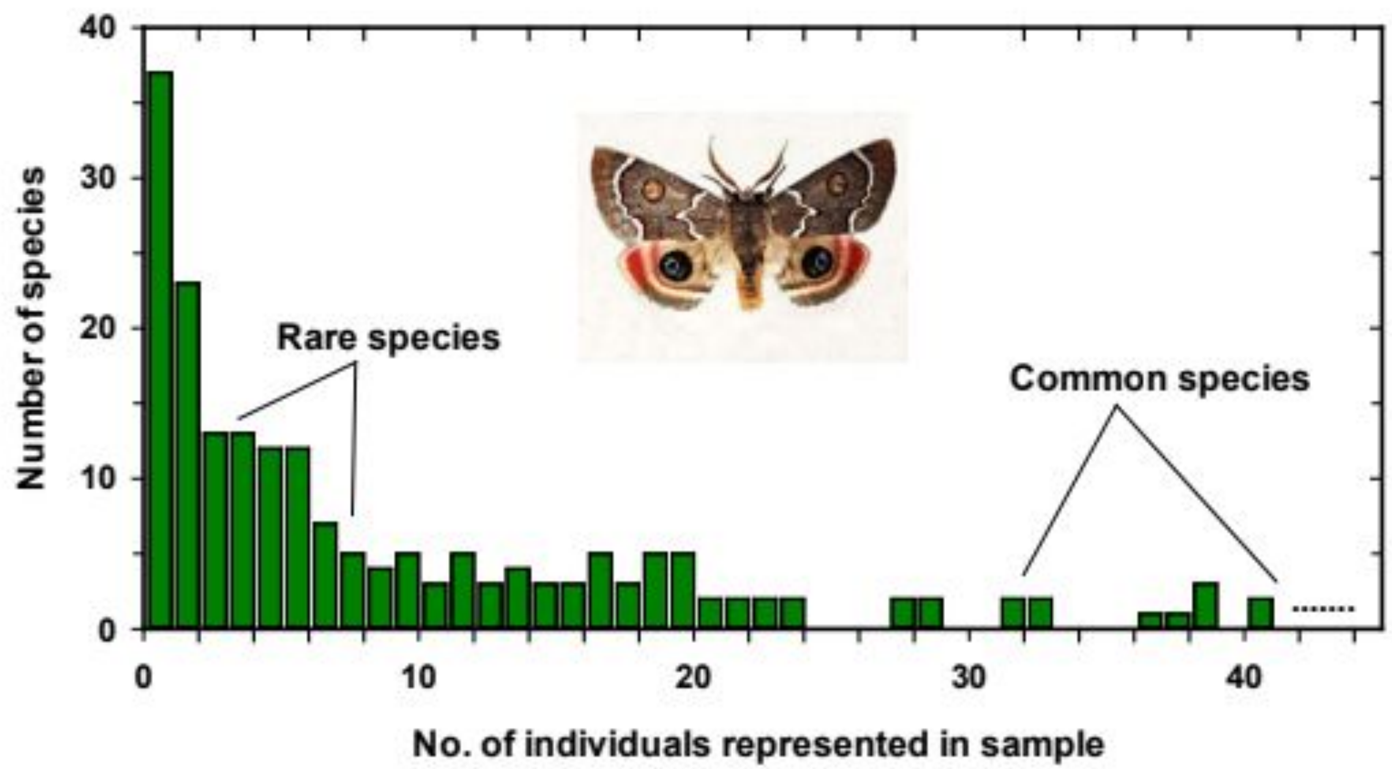


Figure 13.6 Relative abundance of Lepidoptera (butterflies and moths) captured in a light trap at Rothamsted, England, in 1935. Not all of the abundant species are shown. There were 37 species represented in the catch by only a single specimen (rare species); one very common species was represented by 1799 individuals in the catch (off the graph to the right!). A total of 6814 individuals were caught, representing 197 species. Six common species made up 50 percent of the total catch. (*Source:* Williams, 1964.)

Abundância de Espécies em Comunidades tem padrão log-normal

Diversidade de Espécies

- Função de riqueza, heterogeneidade e equitabilidade
- Estes componentes podem ser analisados separadamente ou em conjunto na forma de índices
- Porque alguns lugares tem muitas espécies e outros poucas espécies? Vários fatores
- Interações abióticas e bióticas podem determinar diversidade

Índices Diversidade

be measured by the Shannon-Wiener function[*]:

$$H' = -\sum_{i=1}^{s}(p_i)(\log_2 p_i) \tag{13.36}$$

where:

H' = Information content of sample (bits/individual)

= Index of species diversity

s = Number of species

p_i = Proportion of total sample belonging to ith species

Chao and Shen (2003) adjusted the traditional Shannon-Wiener index to account for the fact that some species are missed in the sample, and they suggested an improved Shannon index:

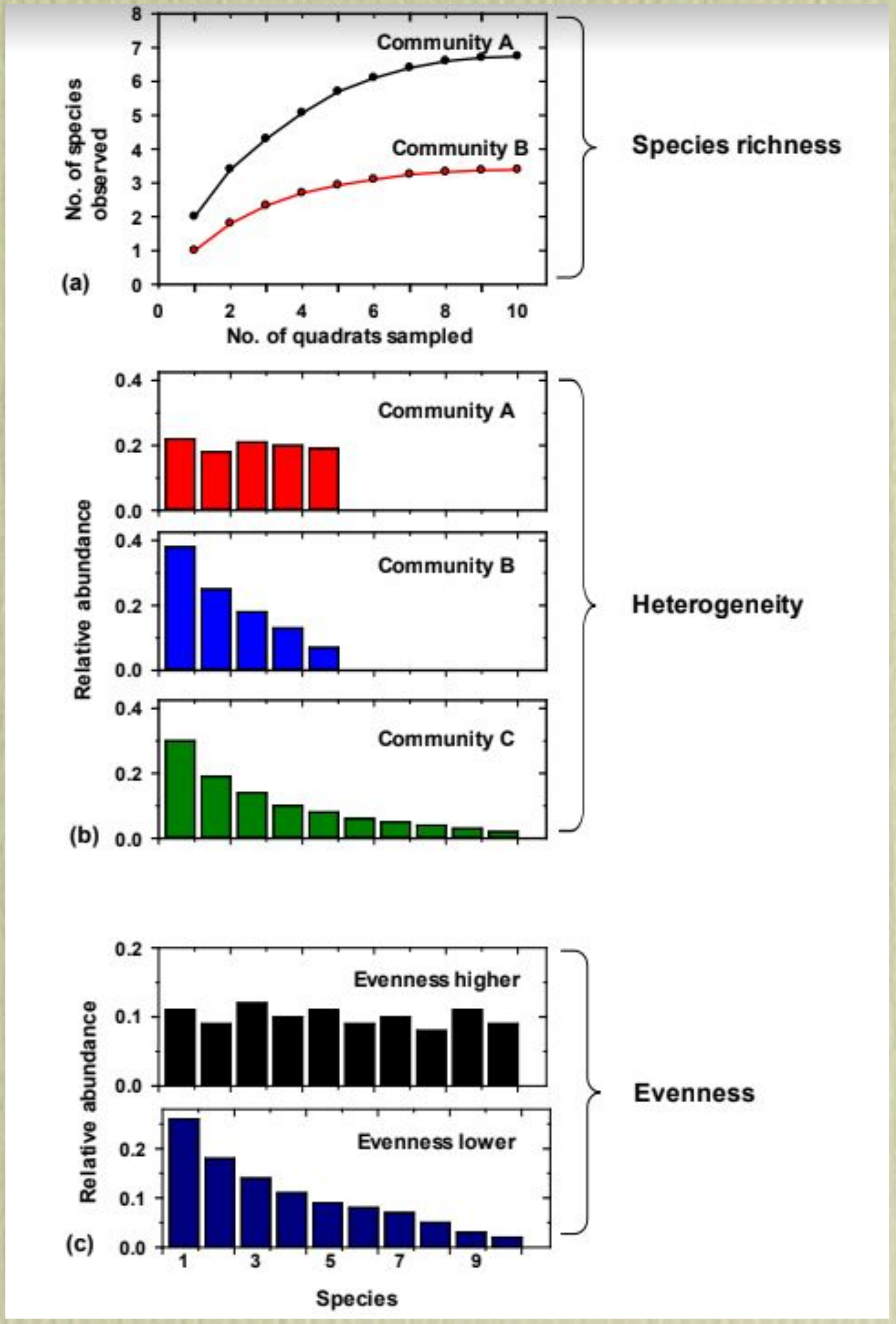
$$\hat{H}' = -\sum_{i=1}^{s}\frac{\hat{p}_i \log_2(\hat{p}_i)}{\left(1-\left[1-\hat{p}_i\right]^n\right)} \tag{13.37}$$

where all variables are defined above and n = total sample size

Combina dados de riqueza e abundância

Componentes da Diversidade

- Riqueza de Espécies
- Heterogeneidade (riqueza vs homogeneidade)
- Homogeneidade (abundâncias similares ou diferentes)



Determinantes de Colonização

- O número de espécies locais vai depender de 5 variáveis: diversidade regional, colonização, hábitats, nichos, interações bióticas

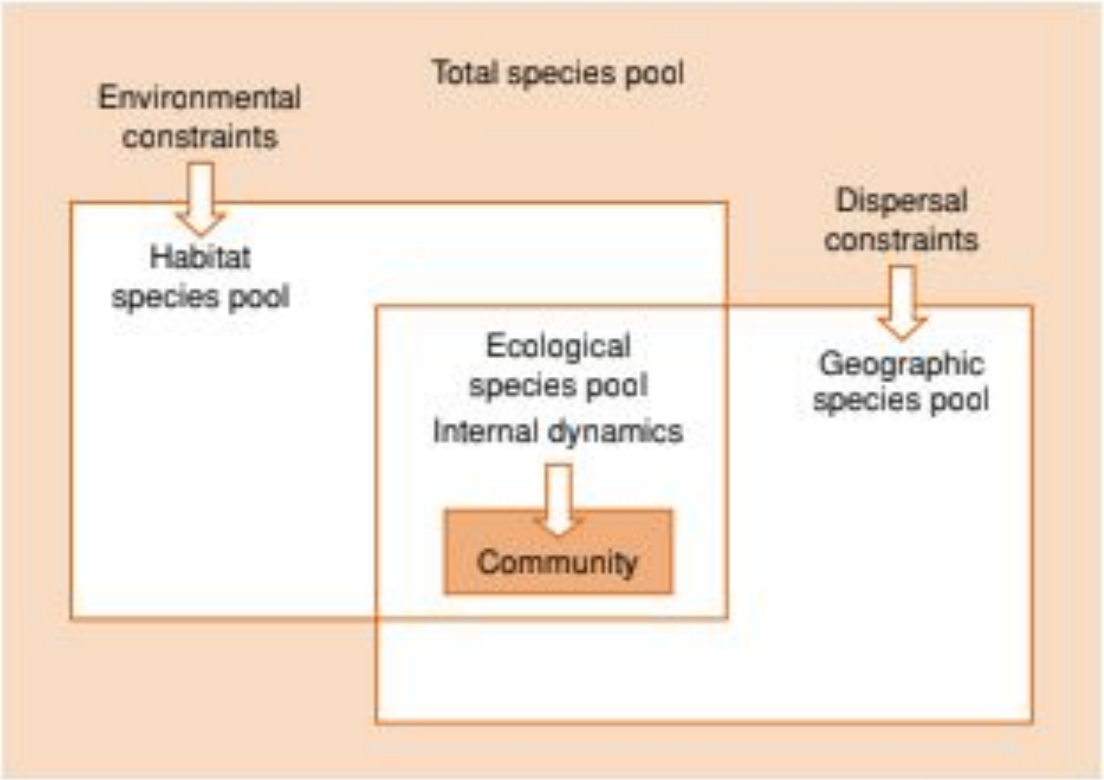
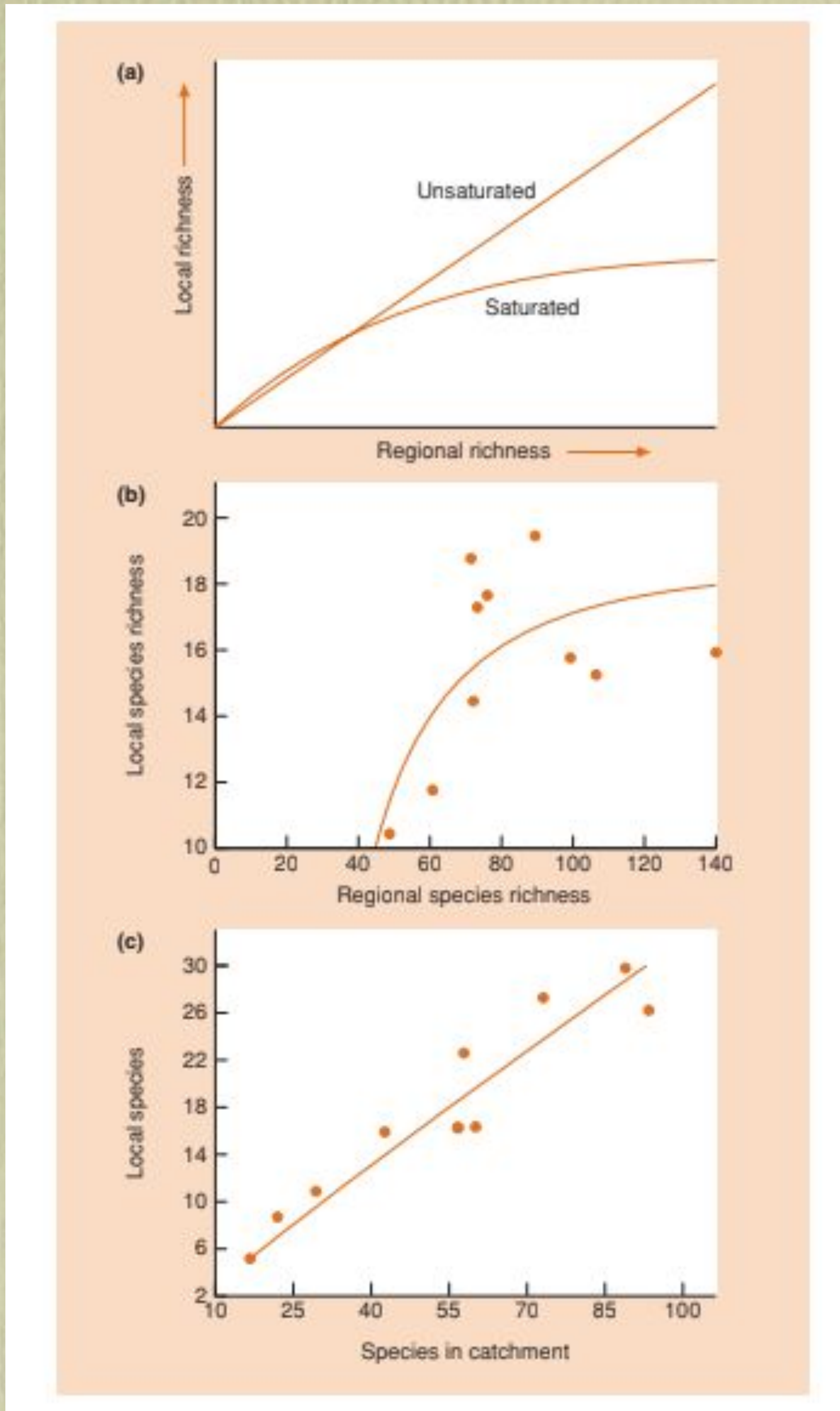


Figure 16.1 The relationships among five types of species pools: the total pool of species in a region, the geographic pool (species able to arrive at a site), the habitat pool (species able to persist under the abiotic conditions of the site), the ecological pool (the overlapping set of species that can both arrive and persist) and the community (the pool that remains in the face of biotic interactions). (Adapted from Belyea & Lancaster, 1999; Booth & Swanton, 2002.)

Riqueza de Espécies

- Relação Entre Diversidade Regional e Saturação local

- Diversidade Local depende de saturação de hábitats e de espécies disponíveis



Biogeografia de Ilhas

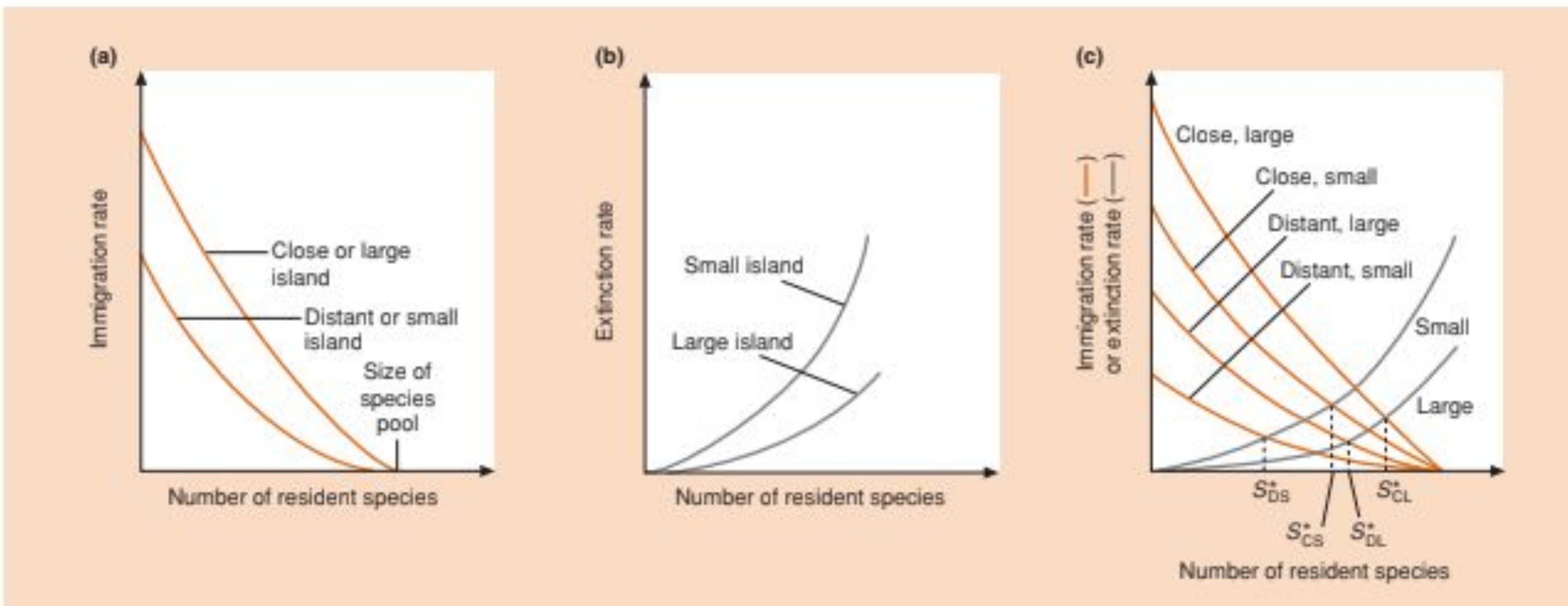


Figure 21.11 MacArthur and Wilson's (1976) equilibrium theory of island biogeography. (a) The rate of species immigration on to an island, plotted against the number of resident species on the island, for large and small islands and for close and distant islands. (b) The rate of species extinction on an island, plotted against the number of resident species on the island, for large and small islands. (c) The balance between immigration and extinction on small and large and on close and distant islands. In each case, S^* is the equilibrium species richness; C, close; D, distant; L, large; S, small.

Efeito de área e isolamento

Área - Tamanho

- Áreas maiores tem maior número de espécies

- Diversidade de hábitats e tamanho suficiente para manter populações

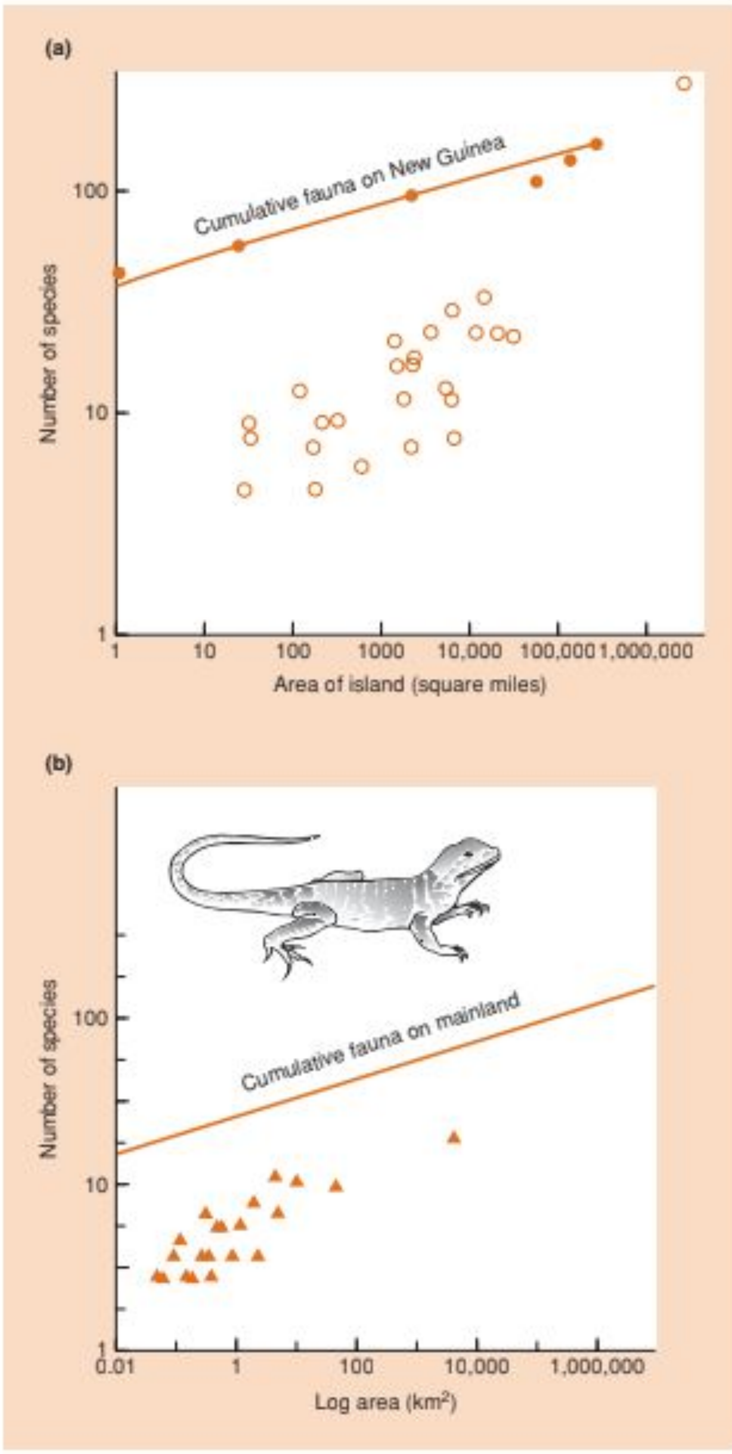


Figure 21.14 (a) The species–area graph for ponerine ants on

Isolamento

- Quanto maior a distância, mais difícil a colonização
- Ilhas isoladas tem menor número de espécies e nichos vazios

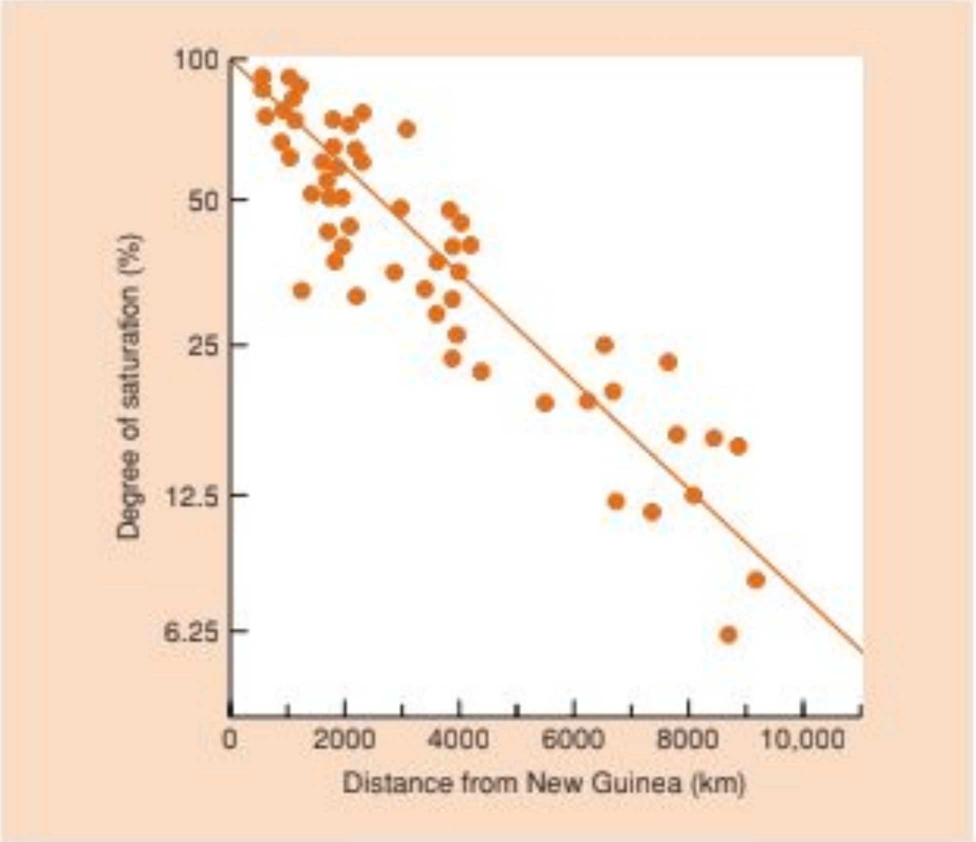


Figure 21.15 The number of resident, nonmarine, lowland bird species on islands more than 500 km from the larger source island of New Guinea expressed as a proportion of the number of species on an island of equivalent area but close to New Guinea, and plotted as a function of island distance from New Guinea. (After Diamond, 1972.)

Isolamento e táxons

- Áreas isoladas tem maior endemismo em táxons com baixo poder de dispersão

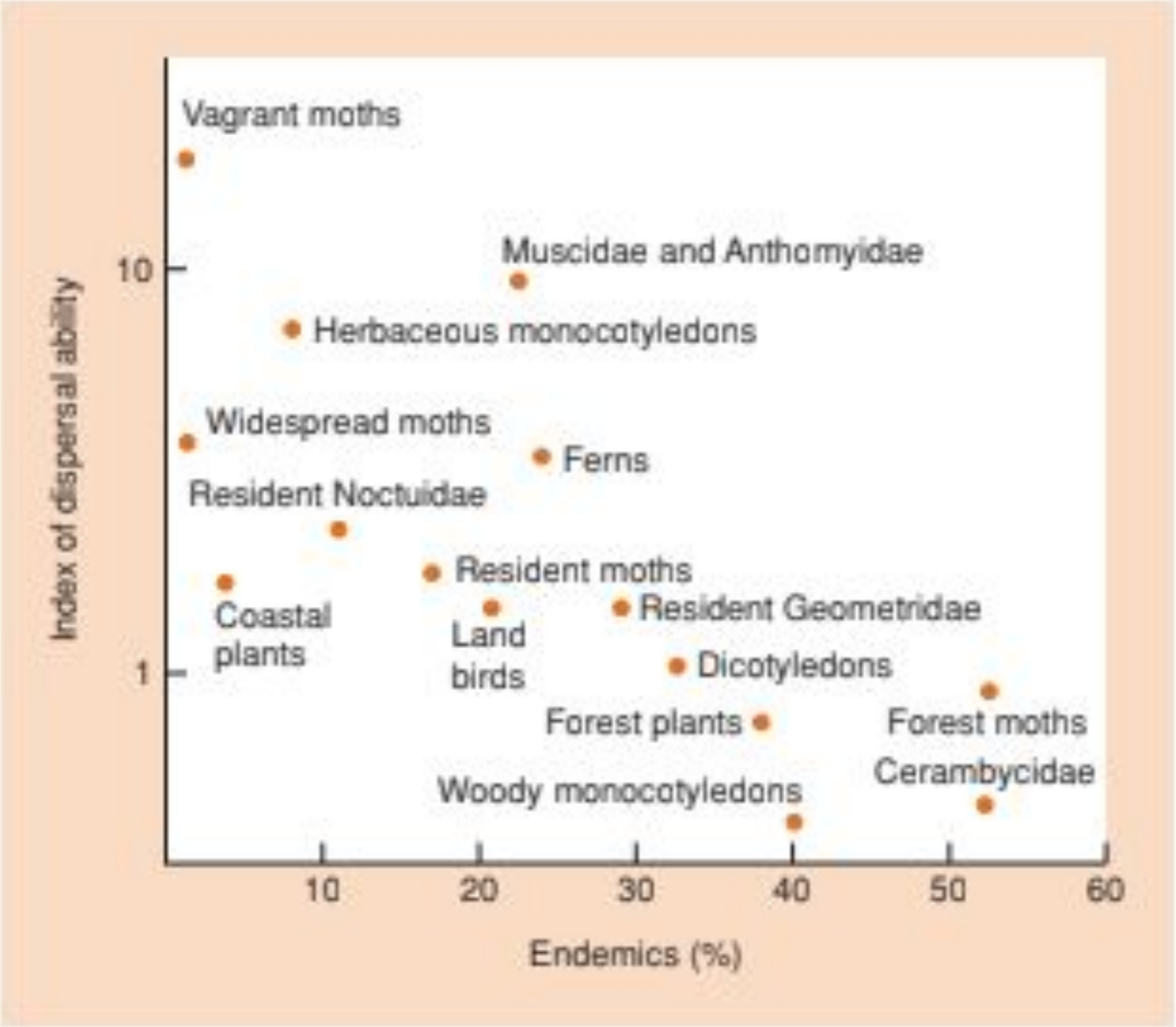


Figure 21.20 Poorly dispersing groups on Norfolk Island have a higher proportion of endemic species, and are more likely to contain species that have reached Norfolk Island from either New Caledonia or New Zealand than species from Australia, which is further away. The converse holds for good dispersers. (After Holloway, 1977.)

Diversidade vs Evapotransp

- Evapotranspiração é indicador de produtividade primária
- Relação positiva e depois estabiliza

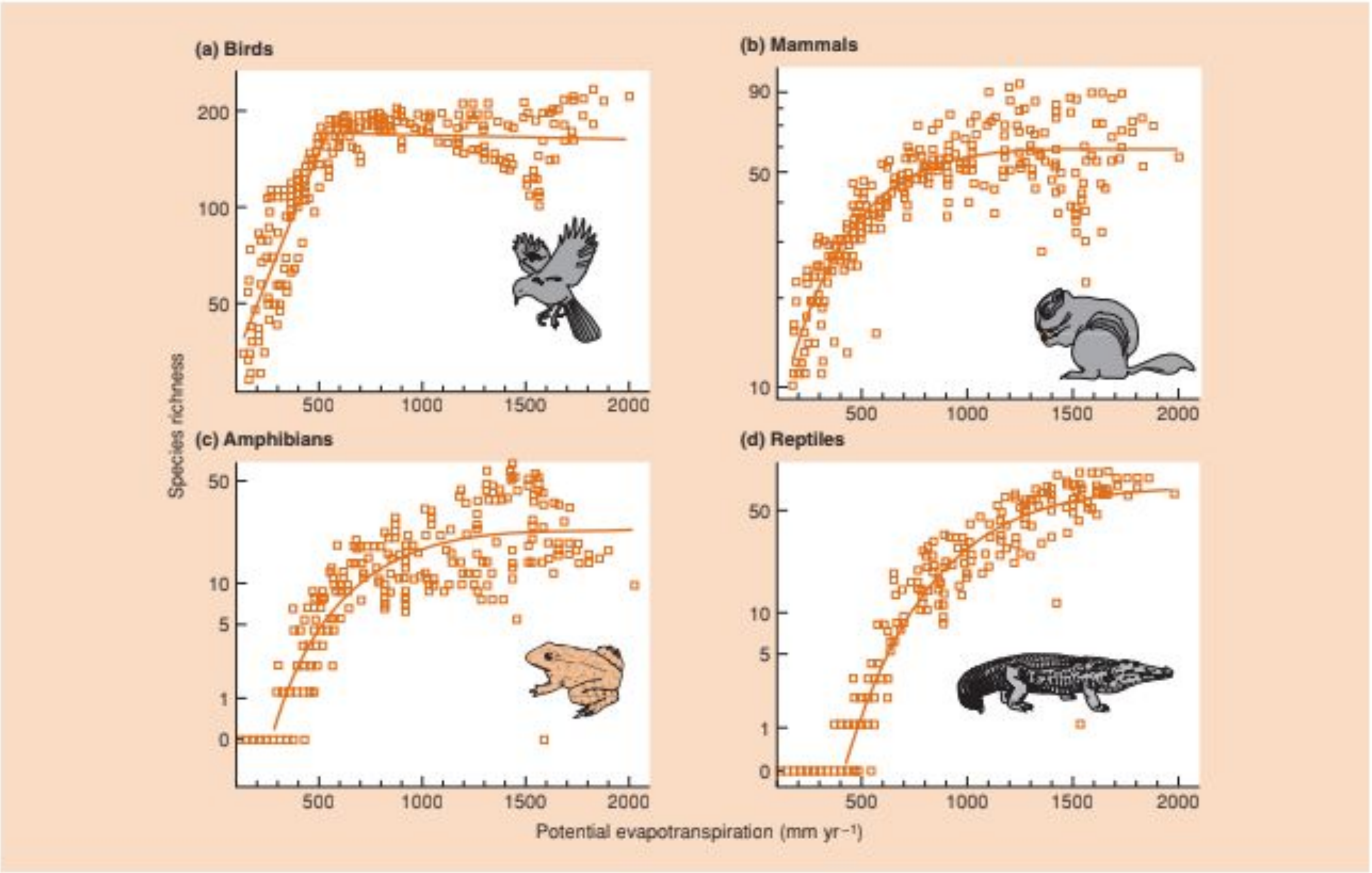


Figure 21.4 Species richness of (a) birds, (b) mammals, (c) amphibians, and (d) reptiles in North America in relation to potential evapotranspiration. (After Currie, 1991.)

Variabilidade Ambiental

- Alta variabilidade ambiental resulta em menores níveis de diversidade. Menor número de espécies se adaptam a grandes variações

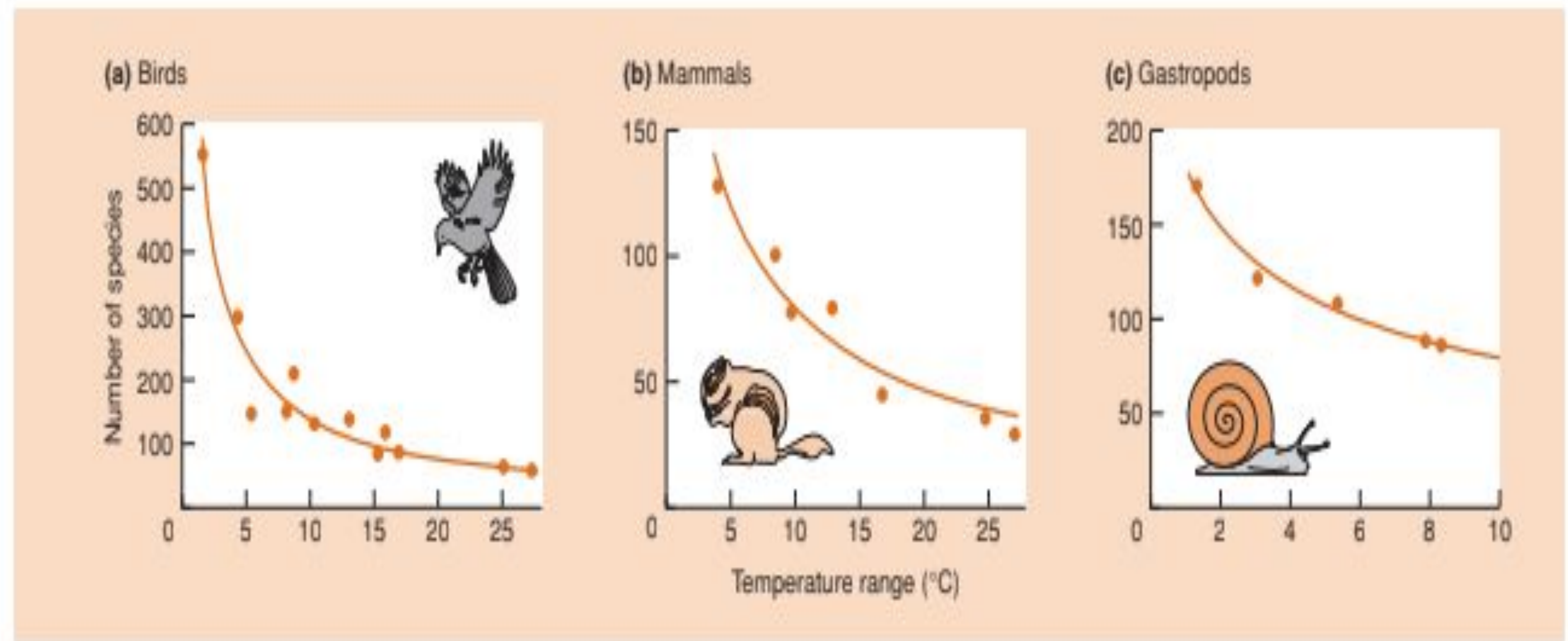
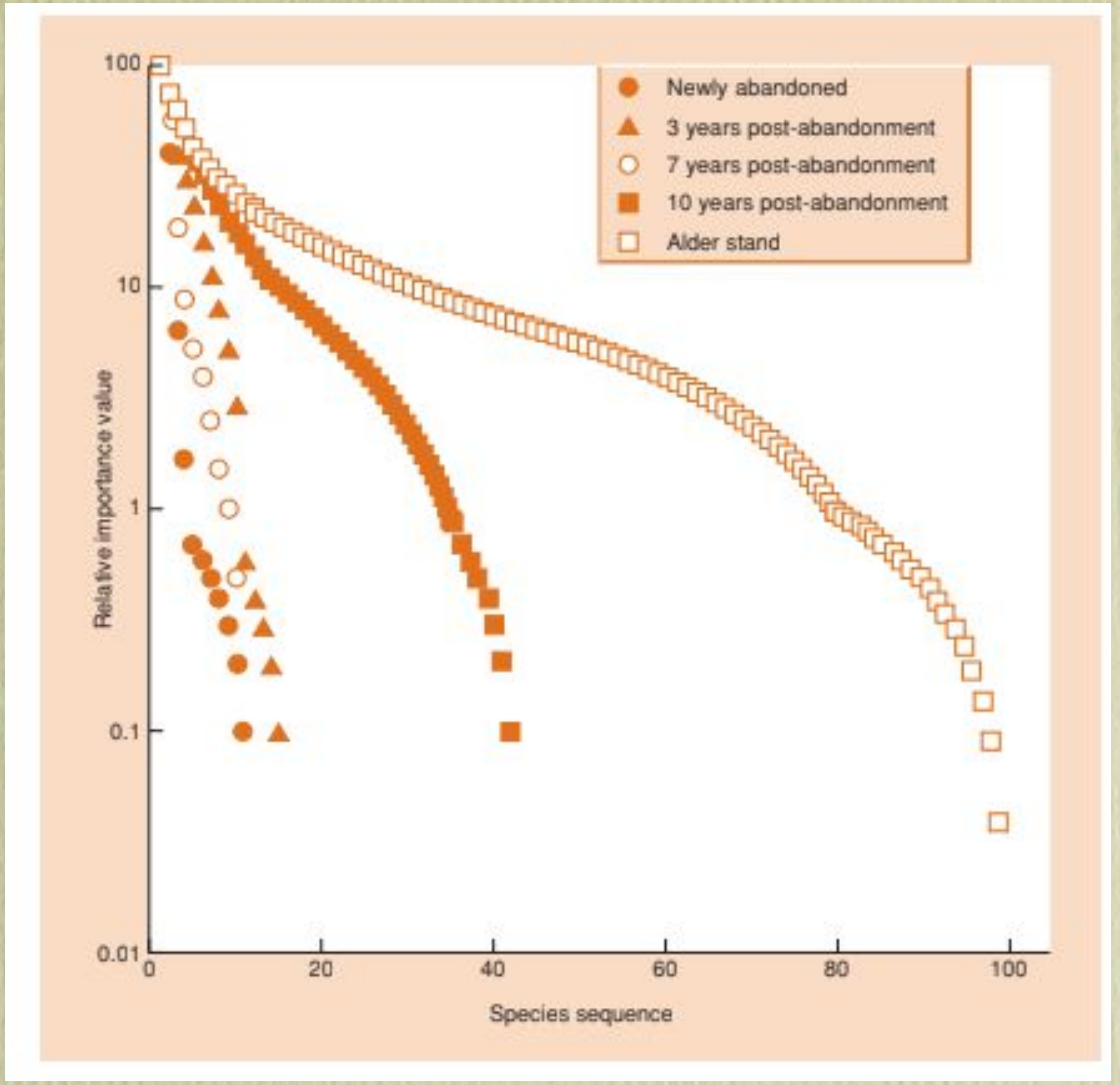


Figure 21.9 Relationships between species richness and the range of monthly mean temperatures at sites along the west coast of North America for (a) birds, (b) mammals and (c) gastropods. (After MacArthur, 1975.)

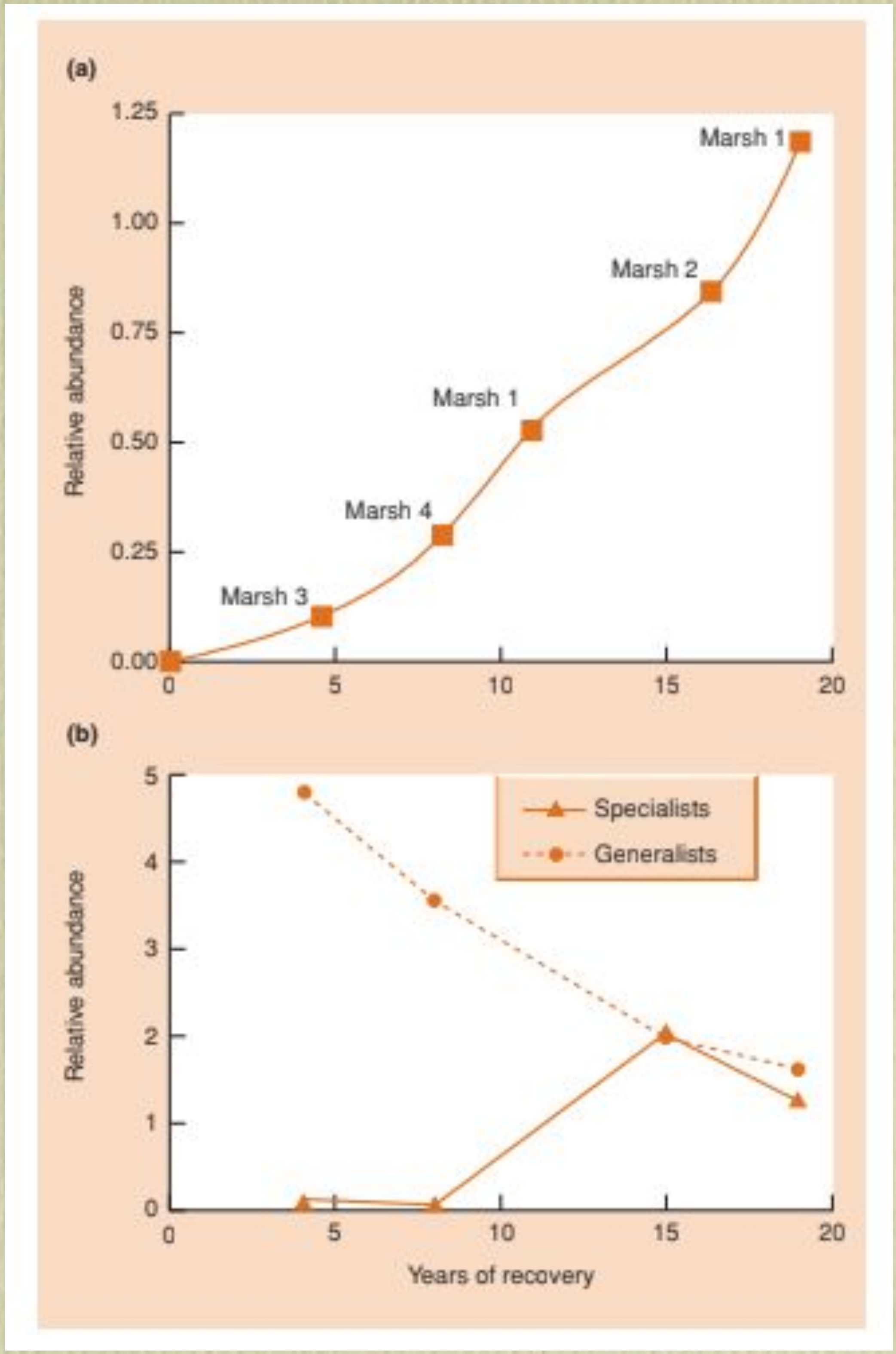
Dominância e Sucessão

- Com o tempo o número de espécies aumenta e há maior equitabilidade



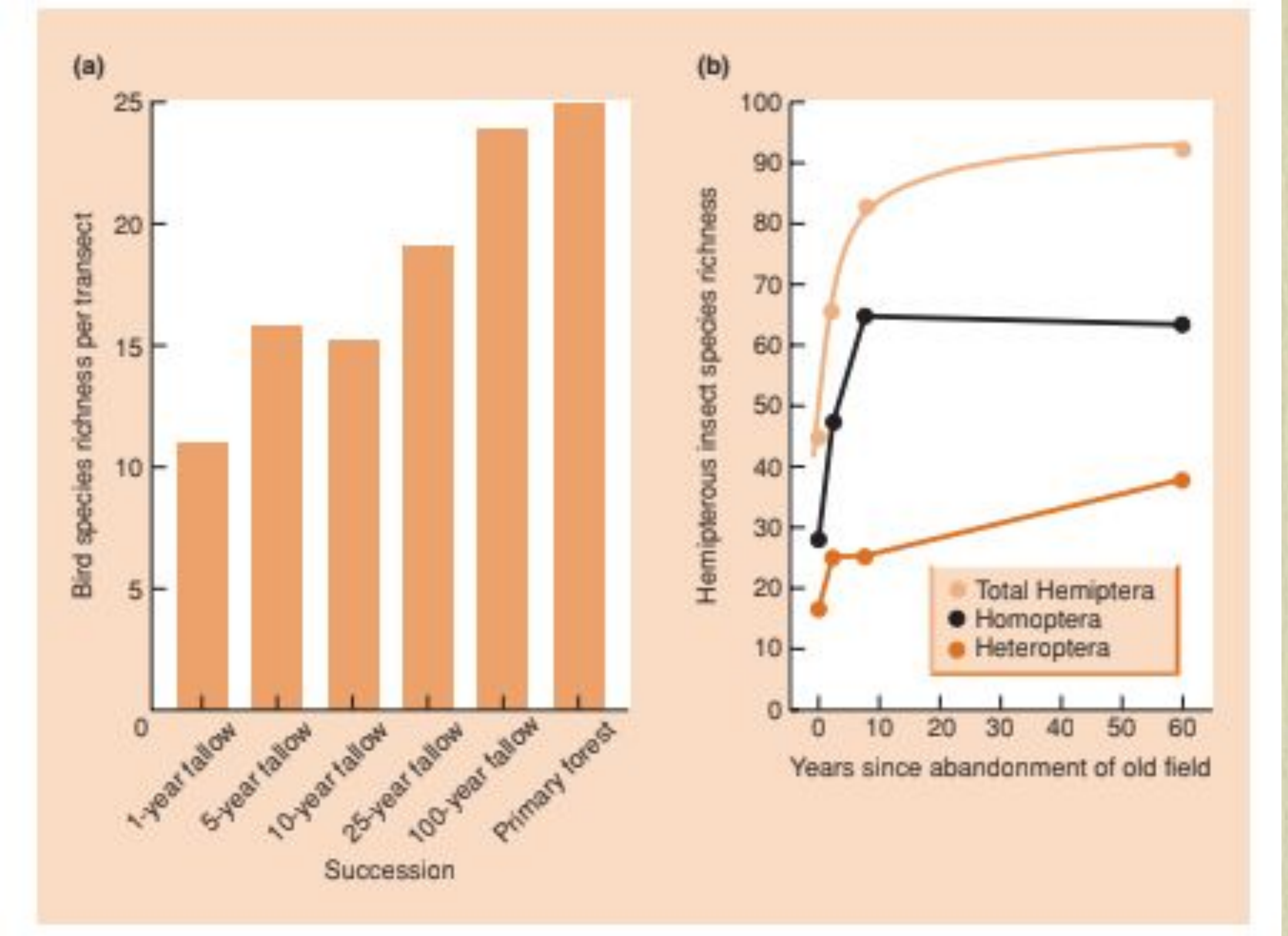
Nicho Ecológico e Sucessão

- Com o tempo a proporção de especialistas aumenta em relação aos generalistas



Riqueza e Sucessão

- Riqueza de Espécies aumenta com o tempo



Hábitats vs Nichos

- A granularidade dos hábitats e micro-hábitats permite a caracterização de nichos e a coexistência

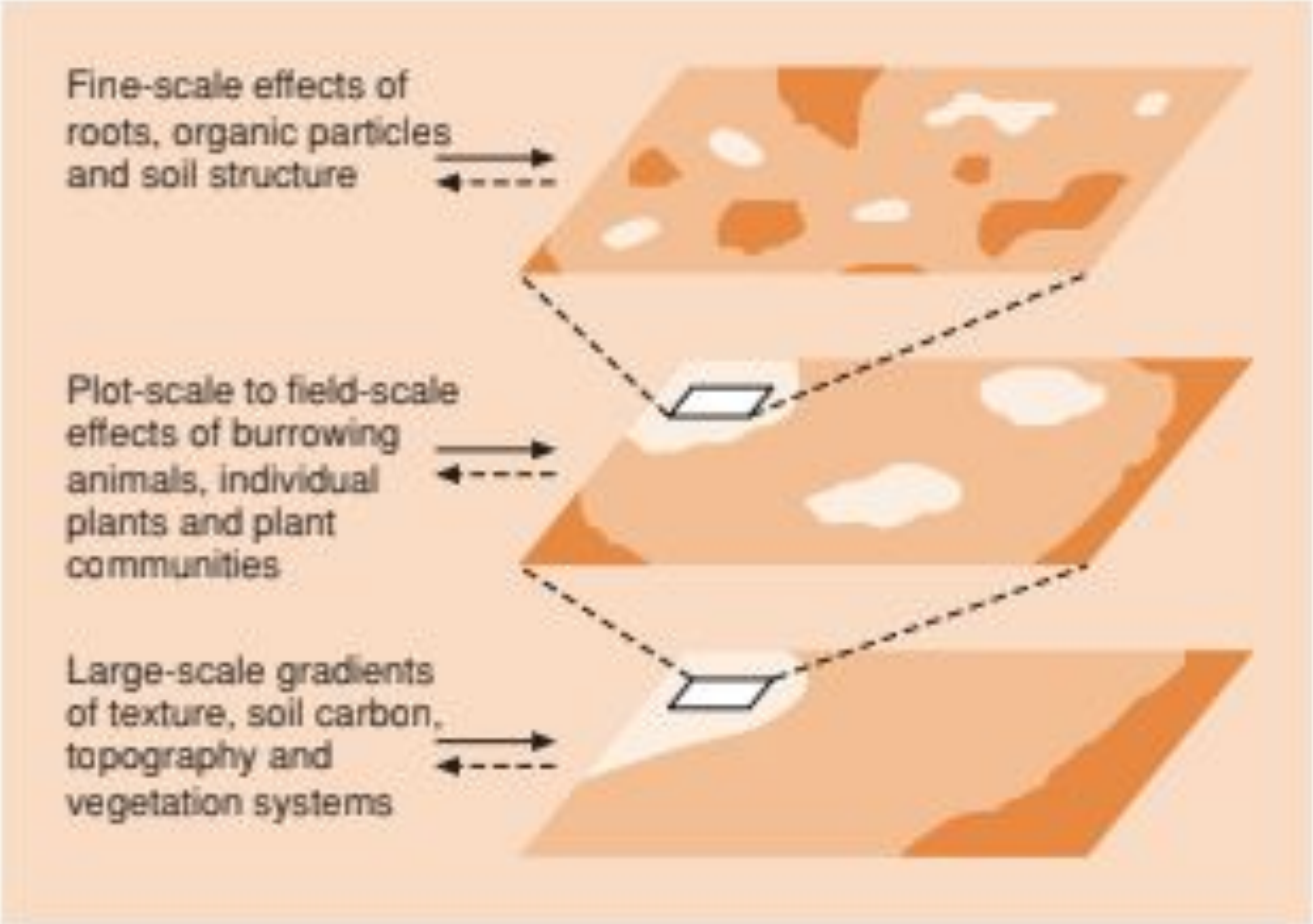


Figure 16.8 Determinants of spatial heterogeneity of communities of soil organisms including bacteria, fungi, nematodes, mites and collembolans. (After Ettema & Wardle, 2002.)

Heterogeneidade e Nichos

- Dimensões de nichos ecológicos

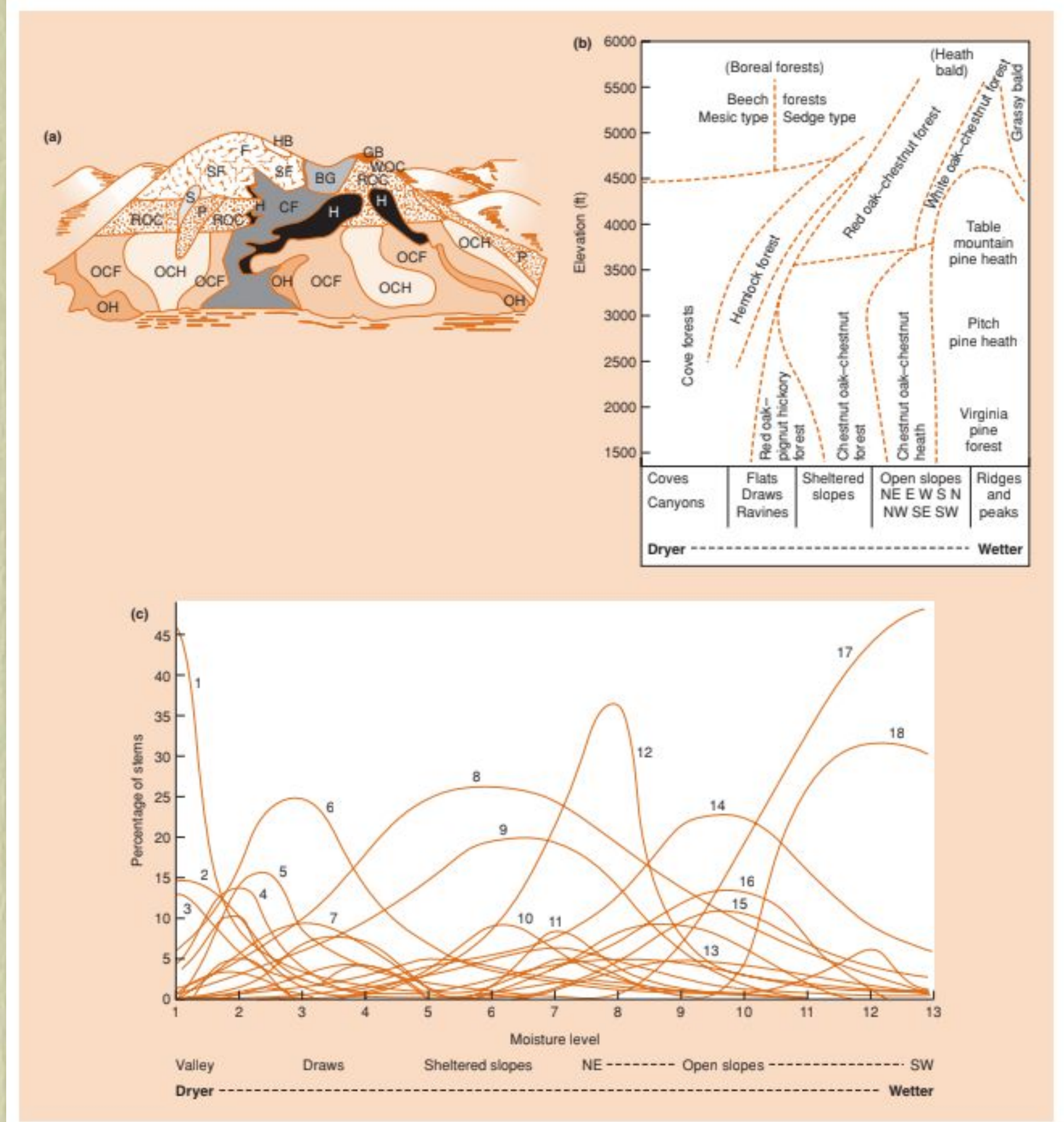
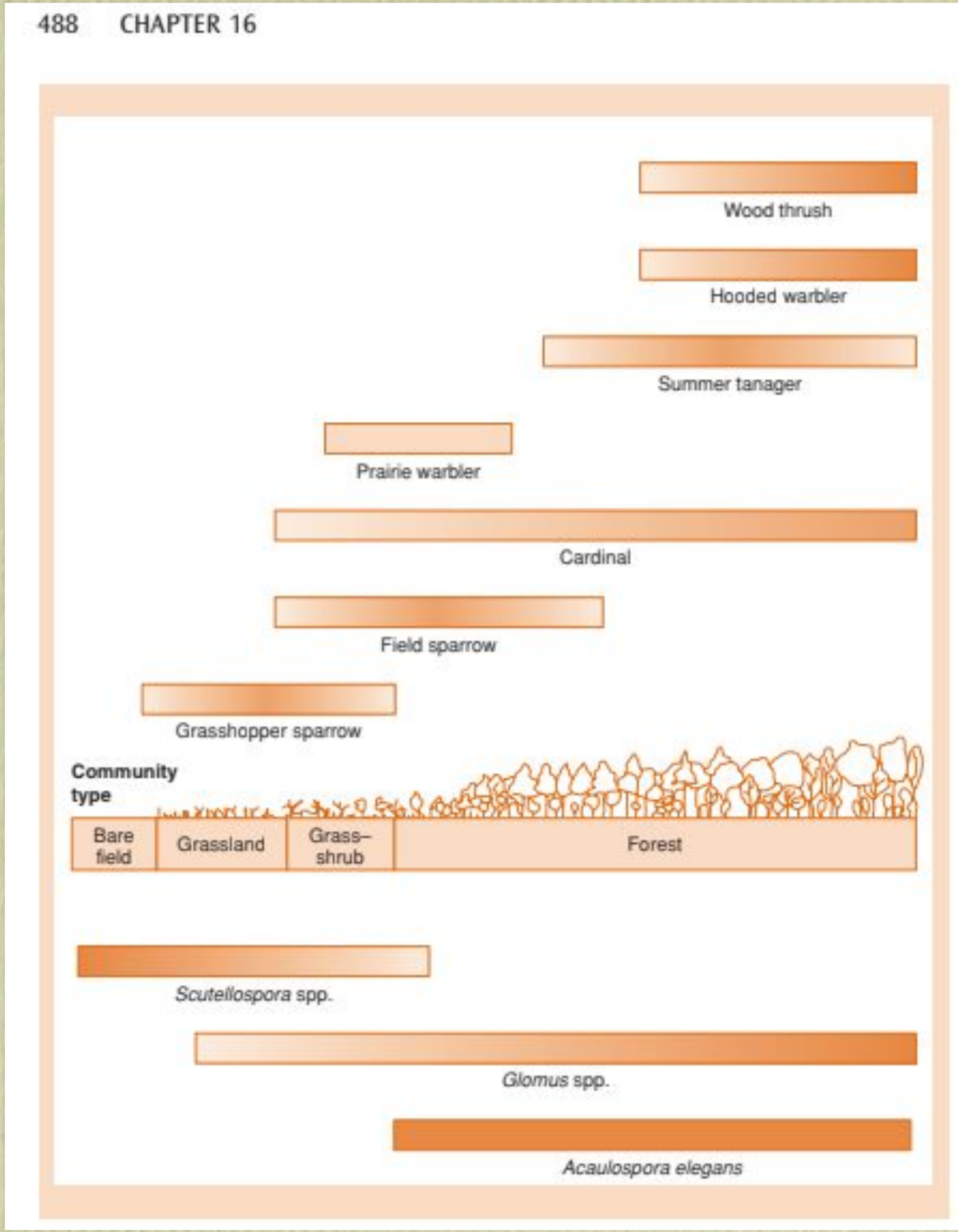


Figure 16.6 Three contrasting descriptions of distributions of the characteristic dominant tree species of the Great Smoky Mountains, Tennessee. (a) Topographic distribution of vegetation types on an idealized west-facing mountain and valley. (b) Idealized graphic arrangement of vegetation types according to elevation and aspect. (c) Distributions of individual tree populations (percentage of stems present) along the moisture gradient. Vegetation types: BG, beech gap; CF, cove forest; F, Fraser fir forest; GB, grassy bald; H, hemlock

Seleção de Hábitats

- As interações bióticas constroem hábitats a partir de variáveis ambientais e ecossistemas



Perturbação e Diversidade

- As perturbações criam ambientes efêmeros e aumentam a diversidade local. Altos níveis de perturbação entretanto favorecem extinções levando a perda de diversidade

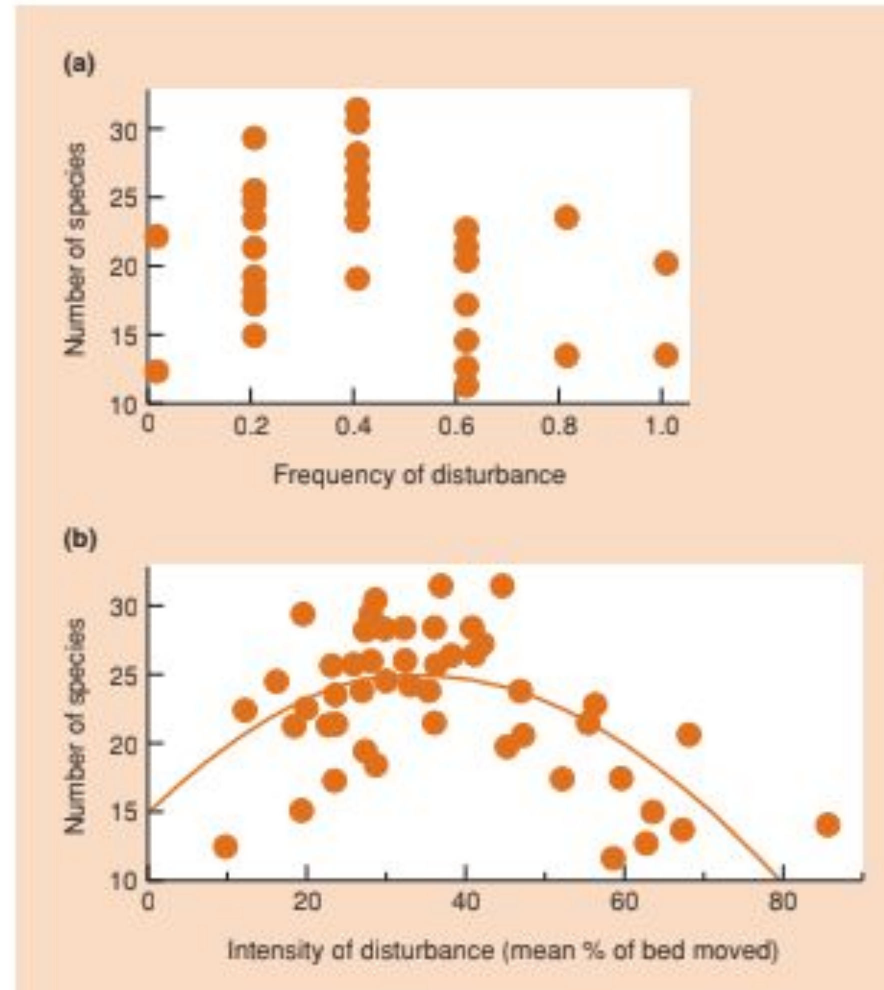
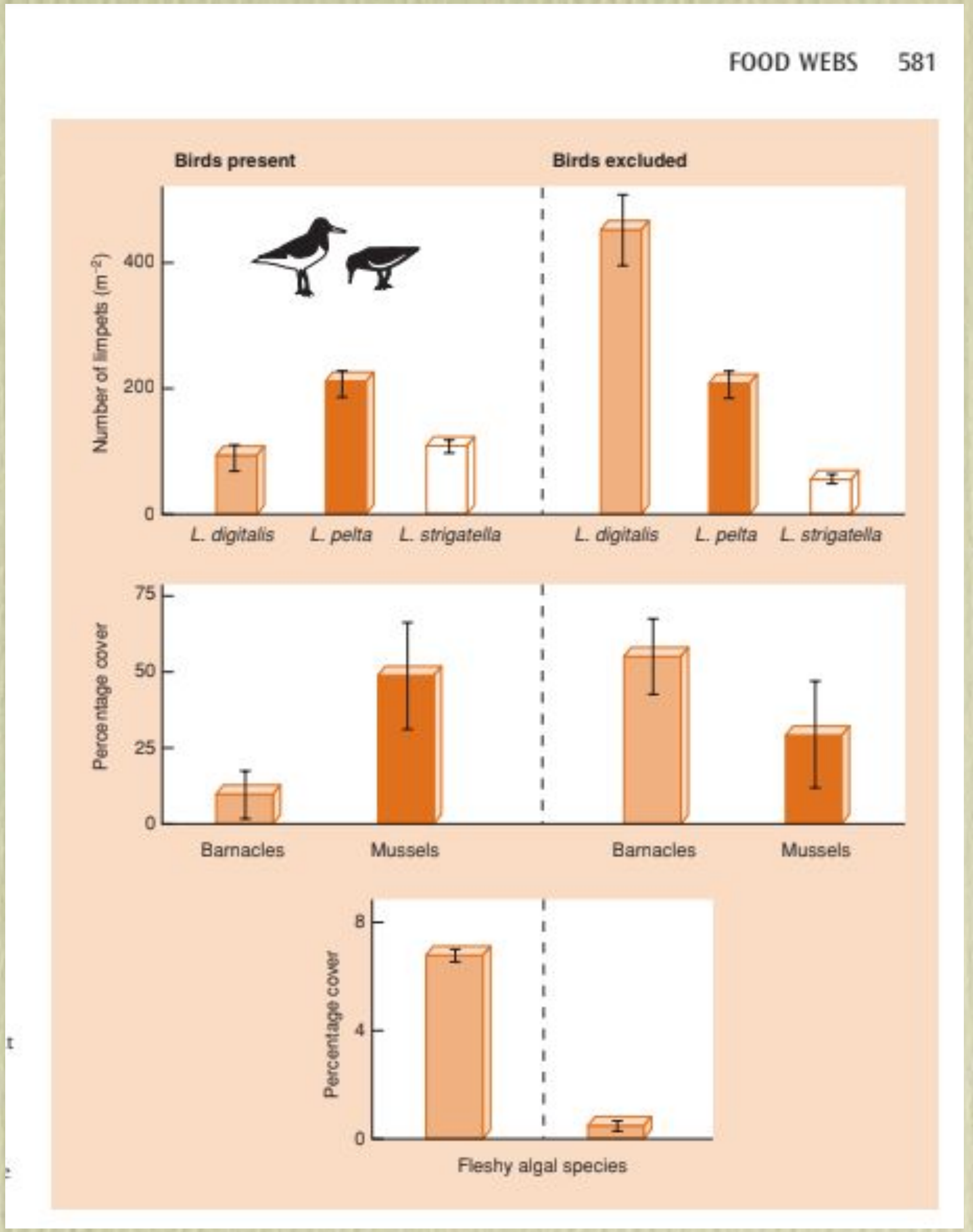


Figure 16.18 Relationship between invertebrate species richness and: (a) frequency of disturbance – assessed as the number of occasions in 1 year when more than 40% of the bed moved (analysis of variance significant at $P < 0.0001$), and (b) intensity of disturbance – average percentage of the bed that moved (polynomial regression fitted, relationship significant at $P < 0.001$) assessed at 54 stream sites in the Taieri River, New Zealand. The patterns are essentially the same; intensity and frequency of disturbance are strongly correlated. (After Townsend *et al.*, 1997.)

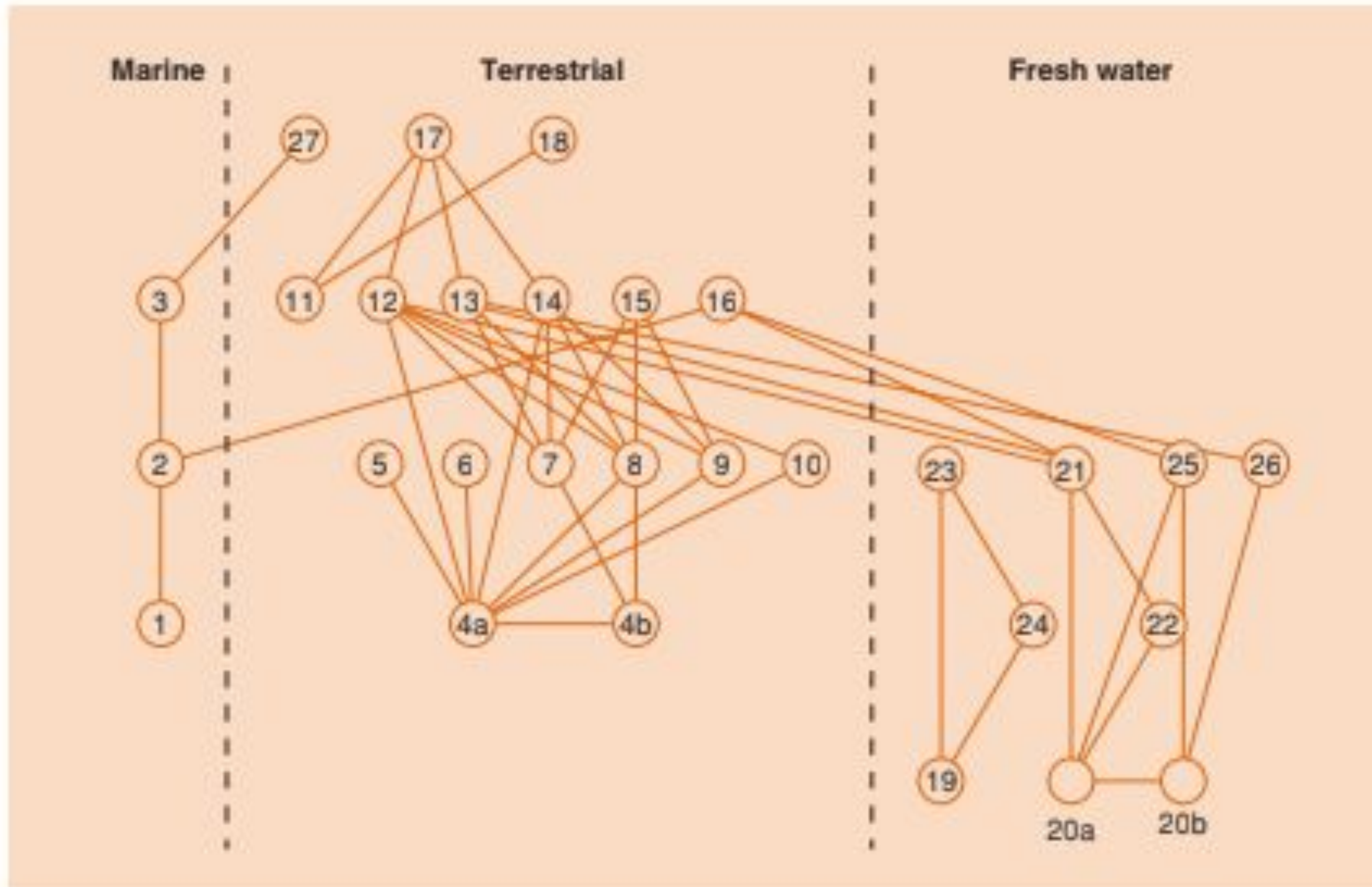
Predação e Diversidade

- A presença de predadores pode reduzir populações de presas e reduzir competição entre estas espécies, levando a maior coexistência e diversidade da comunidade



Redes Tróficas

Figure 20.18 The major interactions within and between three interconnected habitats on Bear Island in the Arctic Ocean. 1, plankton; 2, marine animals; 3, seals; 4a, plants; 4b, dead plants; 5, worms; 6, geese; 7, Collembola; 8, Diptera; 9, mites; 10, Hymenoptera; 11, seabirds; 12, snow bunting; 13, purple sandpiper; 14, ptarmigan; 15, spiders; 16, ducks and divers; 17, Arctic fox; 18, skua and glaucous gull; 19, planktonic algae; 20a, benthic algae; 20b, decaying matter; 21, protozoa a; 22, protozoa b; 23, invertebrates a; 24, Diptera; 25, invertebrates b; 26, microcrustacean; 27, polar bear. (After Pimm & Lawton, 1980.)



Redes Tróficas Dentro e Entre Ecossistemas

Diversidade Urbana

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CrossMark

Determinants of species richness within and across taxonomic groups in urban green spaces

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Discussion

We found that patch area and habitat heterogeneity (measured as number and diversity of habitat types) were the most important explanatory variables for plants (total, native, and endangered), birds (total and native), and overall species richness. This emphasizes the relevance of the species-area effects and the effects of habitat structure on species richness in urban green spaces. As patch area and

Áreas VerdesUrbanas: diversidade é função de tamanho da área e da diversidade de hábitats

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Landscape and Urban Planning 81 (2007) 193–199
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Invertebrate conservation in urban areas: Ants in the Brazilian Cerrado

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Abstract

We evaluated the potential of different types of urban green spaces for conservation of ground-dwelling ants. The study area, which included 12 public squares, two urban parks, and three natural reserves, is within the highly threatened *Cerrado* biome of central Brazil. We compared ant species richness and composition among the different types of urban green spaces, and evaluated how ant communities are affected by the structural characteristics of these spaces. The results indicate that public squares in commercial areas presented the least conservation value given the predominance of exotic ant species (mostly *Pheidole megacephala*). Public squares in residential areas ranked second, given the relative scarcity of exotic species and the greater richness of native species in these locations compared to those in more urbanized areas. The greatest conservation value was achieved by urban parks, which supported the greatest number of native species and were apparently free of exotic species. The closer proximity of urban parks to other natural areas and the presence of native vegetation within the boundaries of those parks appear to be key characteristics for the existence and maintenance of a relatively diverse ant fauna.

Parques ecológicos urbanos tem a maior diversidade e sem espécies exóticas

Diversidade Urbana

Environmental Pollution 159 (2011) 1974–1983

Contents lists available at ScienceDirect

Environmental Pollution

journal homepage: www.elsevier.com/locate/envpol

Review

Novel urban ecosystems, biodiversity, and conservation

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ABSTRACT

With increasing urbanization the importance of cities for biodiversity conservation grows. This paper reviews the ways in which biodiversity is affected by urbanization and discusses the consequences of different conservation approaches. Cities can be richer in plant species, including in native species, than rural areas. Alien species can lead to both homogenization and differentiation among urban regions. Urban habitats can harbor self-sustaining populations of rare and endangered native species, but cannot replace the complete functionality of (semi-)natural remnants. While many conservation approaches tend to focus on such relict habitats and native species in urban settings, this paper argues for a paradigm shift towards considering the whole range of urban ecosystems. Although conservation attitudes may be challenged by the novelty of some urban ecosystems, which are often linked to high numbers of nonnative species, it is promising to consider their associated ecosystem services, social benefits, and possible contribution to biodiversity conservation.

Além das áreas verdes é importante considerar todos os ecossistemas urbanos

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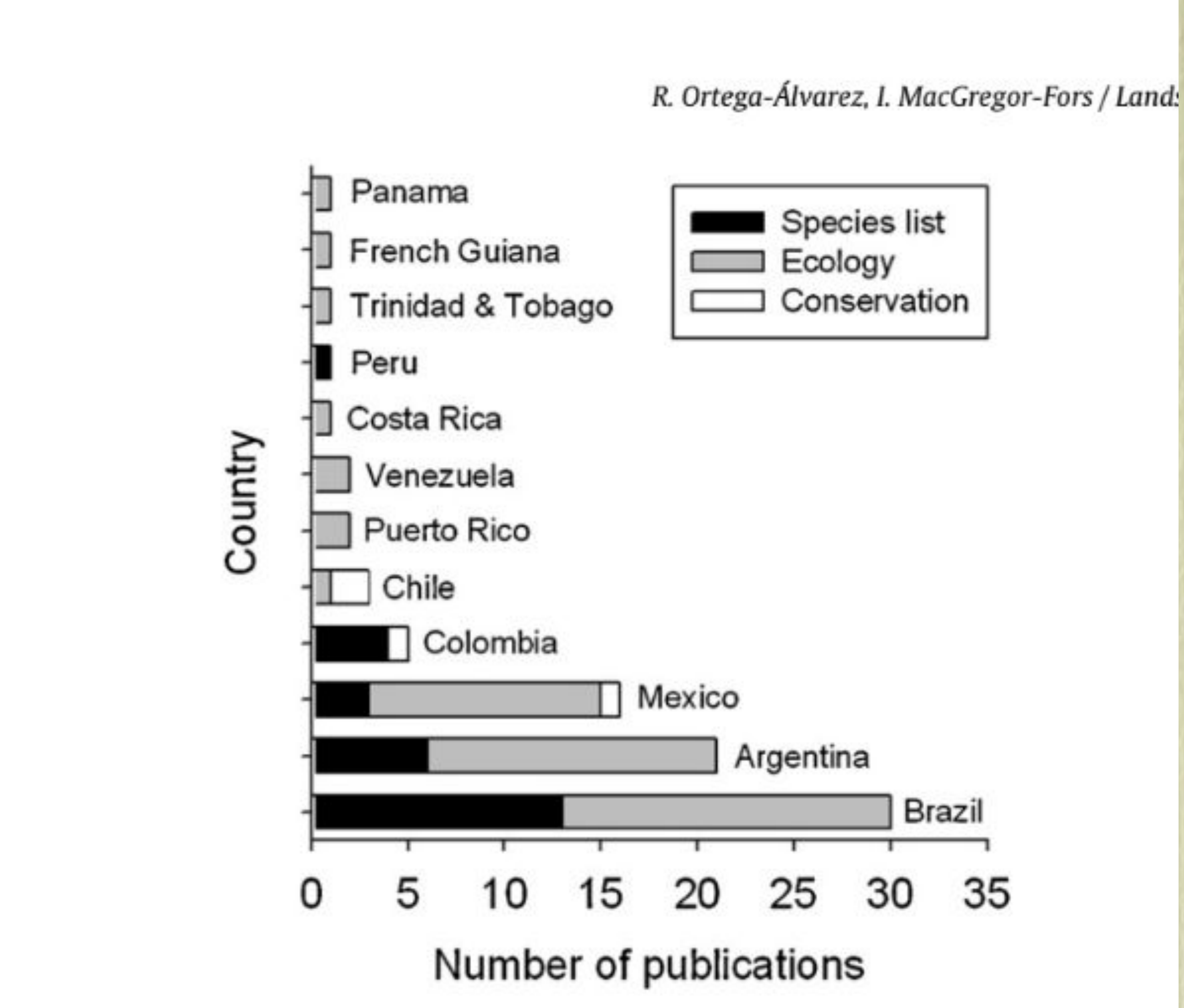

Fig. 1. Number of published studies on urban ornithology in Latin America by country and main subject area.

Brasil lidera em pesquisas sobre avifauna em áreas urbanas

Diversidade Urbana

Urban Ecosystems
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Urbanization homogenizes the interactions of plant-frugivore bird networks

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Abstract
Anthropogenic activities are the main cause of habitat loss and fragmentation, which directly affects biodiversity. Disruption in landscape connectivity among populations may affect complex interactions between species and ecosystem functions, such as pollination and seed dispersal, and ultimately result in secondary extinctions. Urbanization, one of the most intense forms of landscapes changes, has been reported to negatively affect bird and plant diversity. Still, little is known about the effects of urban landscapes on interaction networks. We investigated the relationship between urban landscape structure and plant-frugivore networks at different spatial scales. Coupling interaction data from urban areas and a model selection approach, we evaluated which landscape factors best explained the variation in urban networks properties. Our results indicate that urbanization decreases bird richness, mainly through the loss of habitat specialist species, which results in networks being composed mainly of birds well adapted to urban dwelling. We found that interaction evenness, a measure of homogeneity of interaction distribution between species, increases with urbanization. This is due to the strong dominance that generalist birds had in network composition because they foraged on all available fruits, including exotic plants. The ensuing homogenization of interactions can reduce the resilience of networks and affect the efficiency of ecosystems functions. Thus, urbanization plans should consider the proportion and distribution of green areas within cities, coupling human and ecosystem wellbeing.

Keywords Landscape ecology · Mutualistic network · Urban environmental

Redes tróficas urbanas são diferentes das naturais

Diversidade Urbana

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Factors affecting the occupancy of forest mammals in an urban-forest mosaic in EThekwini Municipality, Durban, South Africa

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ABSTRACT

Urbanisation is one of the most rapidly expanding forms of landscape modification by humans and leads to large-scale loss and fragmentation of native habitat. This can alter the structure, composition and function of remnant habitat. Therefore, understanding the influence of both landscape and patch characteristics is important for understanding factors affecting the distribution of organisms in urbanised landscapes. Consequently, the aim of this study was to establish the responses of forest dwelling mammals to landscape and habitat structure in an urban-forest mosaic in the EThekwini Municipality Area, Durban, South Africa. Using presence and absence data of mammals from camera traps, we modelled occupancy of species using the occupancy modelling framework. The occupancy by *Philantomba monticola* was positively influenced by forest cover (%), woody cover (%), leaf litter (%) and stem density of large trees and negatively influenced by road density. For *Tragelaphus sylvaticus*, *Potamochoerus larvatus* and *Hystix africaeaustralis*, occupancy was influenced positively by forest cover (%), woody cover (%) and foliage height diversity and negatively influenced by road density. For *Genetta tigrina* and

Vegetação e hábitat explicam ocupação dos mamíferos florestais nas áreas urbanas

Diversidade Urbana

31 árvores que você pode plantar em sua calçada, produzem frutos, sombras e flores, não destroem calçadas e nem danificam a rede elétrica

Rua de Caxambu MG com tons diferentes de quaresmeiras e outras especies adequadas para calçadas

As árvores são fundamentais nas ruas e avenidas. Além de embelezar, elas tem um importante papel no equilíbrio térmico, refrescando onde quer que estejam.

Também colaboram com a redução da poluição sonora e do ar, fornecem sombra, refúgio e alimento para as aves. Os benefícios não param por aí, poderíamos falar de fixação de carbono, produção de oxigênio, proteção contra ventos, etc. Mas a escolha da espécie correta é fundamental.

Compatibilizar biodiversidade e uso humano

Diversidade Urbana

LETTER

doi:10.1038/nature11318

Averting biodiversity collapse in tropical forest protected areas

A list of the authors and their affiliations appears at the end of the paper.

The rapid disruption of tropical forests probably imperils global biodiversity more than any other contemporary phenomenon[1–3]. With deforestation advancing quickly, protected areas are increasingly becoming final refuges for threatened species and natural ecosystem processes. However, many protected areas in the tropics are themselves vulnerable to human encroachment and other environmental stresses[4–9]. As pressures mount, it is vital to know whether existing reserves can sustain their biodiversity. A critical constraint in addressing this question has been that data describing a broad array of biodiversity groups have been unavailable for a sufficiently large and representative sample of reserves. Here we present a uniquely comprehensive data set on changes over the past 20 to 30 years in 31 functional groups of species and 21 potential drivers of environmental change, for 60 protected areas stratified across the world's major tropical regions. Our analysis reveals great variation in reserve 'health': about half of all reserves have been effective or performed passably, but the rest are experiencing an erosion of biodiversity that is often alarmingly widespread taxonomically

(mean ± s.d., 19.1 ± 9.6 years) at each protected area. Each interviewed researcher completed a detailed 10-page questionnaire, augmented by a telephone or face-to-face interview (see Supplementary Information). The questionnaires focused on longer-term (approximately 20–30-year) changes in the abundance of 31 animal and plant guilds (trophically or functionally similar groups of organisms), which collectively have diverse and fundamental roles in forest ecosystems (Table 1). We also recorded data on 21 potential drivers of environmental change both inside each reserve and within a 3-km-wide buffer zone immediately surrounding it (Table 1).

Our sample of protected areas spans 36 nations and represents a geographically stratified and broadly representative selection of sites across the African, American and Asia-Pacific tropics (Supplementary Fig. 1). The reserves ranged from 160 ha to 3.6 million ha in size, but most (85%) exceeded 10,000 ha in area (median = 99,350 ha; lower decile = 7,000 ha; upper decile = 750,000 ha). The protected areas fall under various International Union for Conservation of Nature (IUCN) reserve classifications. Using data from the World Database

Como evitar colapso em áreas naturais?

Diversidade Urbana

Table 1 | The 31 animal and plant guilds, and the 21 environmental drivers assessed both inside and immediately outside each protected area.

Guilds	Potential environmental drivers
Broadly forest-dependent guilds	
Apex predators	Changes in natural-forest cover
Large non-predatory species	Selective logging
Primates	Fires
Opportunistic omnivorous mammals	Hunting
Rodents	Harvests of non-timber forest products
Bats	Illegal mining
Understory insectivorous birds	Roads
Raptorial birds	Automobile traffic
Larger frugivorous birds	Exotic plantations
Larger game birds	Human population density
Lizards and larger reptiles	Livestock grazing
Venomous snakes	Air pollution
Non-venomous snakes	Water pollution
Terrestrial amphibians	Stream sedimentation
Stream-dwelling amphibians	Soil erosion
Freshwater fish	River & stream flows
Dung beetles	Ambient temperature
Army or driver ants	Annual rainfall
Aquatic invertebrates	Drought severity or intensity
Large-seeded old-growth trees	Flooding
Epiphytes	Windstorms
Other functional groups	
Ecological specialists	
Species requiring tree cavities	
Migratory species	
Disturbance-favouring guilds	
Lianas and vines	
Pioneer and generalist trees	
Exotic animal species	
Exotic plant species	
Disease-vectoring invertebrates	
Light-loving butterflies	
Human diseases	

Indicadores de áreas naturais e de áreas antropizadas